Firebolt 9 Crypto API Specification

Document status	APPROVED
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API version	Link	Summary
0.1		First full approved version. Approved in Firebolt Crypto WG on 08 Okt 26

Interpretation

This page is an abstract, language-independent definition of the Firebolt Crypto API. The following terms are used:

	Term	Meaning
1	enum	an instance of a type that can have one of a pre-defined set of values. Typically represented by a string in JSON-RPC, but an integer in C++
2	list of	an ordered collection of values
3	list of... unordered	an unordered collection of values
4	one of	an instance of a primitive type, such as unsigned or string, that can have one of a pre-defined set of values. Not an enum
5	monotonic	a numeric value that is greater than the last instance of that attribute
6	generic error	an error that can be returned by any method call, documented in the Errors section
7	specific error	an error that can be returned by specific methods, documented along with the method

Key

Color	Meaning
Green	Approved - ready for development
Red	Not approved - not ready for development

Types

	Type	Definition
1	unsigned	32-bit unsigned integer
2	double	64-bit double-precision floating point number

Errors

	Class	ID	Error		Notes
1	Generic	-xxxxh	Authentication Failed		Operation not permitted for specific app credentials - example, when an app tries to use another apps credentials
2		-xxxxg	Invalid Parameters		Null or Invalid parameters
3		-xxxxa	Operation failed		Includes any resourcing errors, operation not allowed / supported, any internal or hardware error
4	Specific	-xxxxb	File to be loaded is not available or corrupted		

5		-xxxxc	File name does not represent valid key name		
6		-xxxxd	Invalid key type	xxx	
7		-xxxxe	Invalid key		
8		-xxxxf	Session to be reinitialized		The vault session is invalidated. It needs to be reinitialized before next usage.
9		-xxxxi	Invalid Init Data		Length of IV or nonce supplied as parameter is not equal to the expected length
10		-xxxxj	Data size exceeded		
11		-xxxxk	Key Provision Error		
12		-xxxxl	Too small output buffer		
13		-xxxxm	Sealed Key Cannot Export Error		

Methods

	Module	Method	Parameters	Returns	Specific errors	C++17	JS	Description

1	Crypto	generateKey	• keyParam - enum
			  ◦ AES128
			  ◦ AES256
			  ◦ HMAC128
			  ◦ HMAC160
			  ◦ HMAC256
			• keyUsage - list of enums, unordered - optional
			  ◦ ENCRYPT
			  ◦ DECRYPT
			  ◦ SIGN
			  ◦ VERIFY
			  ◦ DERIVE
			  ◦ WRAP
			  ◦ UNWRAP
				• keyId - unsigned	• -xxxxa - Key Creation Error
					• -xxxxe - Invalid File name
					• -xxxxd - Invalid key type
						Yes	No	Generate symmetric algorithm key file, load it into key store and provide handle for further access. Generated key this way shall be considered local to the application generating it - shall not be shared. Not persisted during this call. To use exportKey in case persistence is required.
								
								If keyUsage is omitted, the platform will assume all keyUsages corresponding to the keyParam.

2	Crypto	generateKeyPair	• algorithm - enum
			  ◦ AUTH_DH
			
			Optional parameter for AUTH_DH algo:
			
			{
			• primePDataOffset - unsigned
			• primePDataLength - unsigned
			• generatorGDataOffset - unsigned
			• generatorGDataLength - unsigned
			}
			
			Optional parameter for RSA algo (not supported, just for ex):
			
			{
			• modulusBits - unsigned
			• publicExponent - unsigned
			}
				• privateKeyId - unsigned
				• publicKeyId - unsigned
					None	Yes	No	Generates an ephemeral or application-scoped Diffie-Hellman public /private key pair for subsequent AUTH_DH derivation.
								
								Mapping from Netflix APIs:
								
								generatorG; // from base_buf
								
								primeP; // from modulus

3	Crypto	loadKey	• keyName - string, locator that points to a key that is pre-provisioned • provisionType - enum - optional ◦ pre-provisioned (global)	• unsigned - keyId (Reference to the loaded key)	• -xxxxb - Not available or corrupted • -xxxxc - Does not represent valid key	Yes	No	Load a pre-provisioned key by locator
4	Crypto	unloadKey	• keyId - unsigned	None	None	Yes	No	Delete key from key store (may mean unload key from TA if used).
5	Crypto	keyExists	• keyName - string	• bool • unsigned - keyId	• -xxxxh - Authentication failure	Yes	No	Returns true if a valid key exists (loaded to be used) with the key name. Returns false otherwise.

6	Crypto	getKeyInfo	• keyId - unsigned	• keyType - enum
				  ◦ PUBLIC
				  ◦ PRIVATE
				  ◦ SYMMETRIC
				• keyUsage -list of enums, unordered
				  ◦ ENCRYPT
				  ◦ DECRYPT
				  ◦ SIGN
				  ◦ VERIFY
				  ◦ DERIVE
				  ◦ WRAP
				  ◦ UNWRAP
				• keyAlgo - enum
				  ◦ AES_CBC
				  ◦ AES_CTR
				  ◦ SHA256
				  ◦ SHA384
				  ◦ SHA512
				  ◦ AUTH_DH
				  ◦ DH
					• -xxxxe - Invalid key	Yes	No	Get key info / size from keystore

7	Crypto	flushKeys	None	None	None	Yes	No	Unloads all the keys loaded until now. They cannot be used again. Returns INVALID_KEY error on further key usage.

8	Crypto	importKey	• keyType - enum
			  ◦ PUBLIC
			  ◦ PRIVATE
			  ◦ SYMMETRIC
			• Import/Export Type
			  ◦ Protected
			  ◦ Clear
			• Key blob to be imported - device-bound, integrity-protected, confidentiality-protected blob
				• keyId - unsigned (Reference to the loaded key)	• -xxxxk - Key Provision Error	Yes	No	Load the exported key material from the same device / platform into the key store.
								
								• Verifies integrity and device binding before creating a key
								• Rejects corrupted or foreign-device blobs
								• Does not create a partially usable key on failure
								• May return a different key ID from the original

9	Crypto	exportKey	• keyId - unsigned
			• Import/Export Type - enum
			  ◦ Protected
			  ◦ Clear
				• keyBlob - list of unsigned, ordered.	• -xxxxm - Sealed Key Cannot Export Error
					• -xxxxe - Invalid key
						Yes	No	Exports the key material for usage by the platform after an import.
								
								Protected export shall preserve key material and all platform key metadata required to restore equivalent key behavior. The returned blob shall be device-bound, integrity-protected, and confidentiality-protected.

10	Crypto	sign	• algorithm - enum
			  ◦ HMAC
			• keyId - unsigned (Reference to the loaded key)
			• digestAlgorithm - enum
			  ◦ SHA-256
			• inputDataOffset - unsigned
			• inputDataLength - unsigned
				• outputDataOffset - unsigned
				• outputDataLength - unsigned
					• -xxxxc / Invalid key
					• -xxxxd / Invalid key type
					• -xxxxf / Session to be reinitialized
					• -xxxxl - Output buffer too small
						Yes	No	Calculate the MAC / signature value and returns.
								
								keyId shall match with a loaded key compatible with the selected algorithm.
								
								Examples:
								
								- HMAC: secret symmetric key

11	Crypto	verify	• algorithm - enum
			  ◦ HMAC
			• keyId - unsigned (Reference to the loaded key)
			• digestAlgorithm - enum
			  ◦ SHA256
			• inputDataOffset - unsigned
			• inputDataLength - unsigned
			• expectedSignatureOffset - unsigned
			• expectedSignatureLength - unsigned
				• bool - verification status	• -xxxxx / Invalid key
					• -xxxxd / Invalid key type
					• -xxxxy / Session to be reinitialized
						Yes	No	Calculate the MAC / signature value and verifies if the expected signature matches. Returns true if verification is successful or returns false.
								
								keyId shall match with a loaded key compatible with the selected algorithm.
								
								Examples:
								
								- HMAC: secret symmetric key

12	Crypto	encrypt	• cipherAlgorithm - enum ◦ AES_CBC • keyId - unsigned (Reference to the loaded key) • initDataOffset - unsigned (optional) • InitDataLength - unsigned (optional) • inputDataOffset - unsigned • inputDataLength - unsigned	• encryptDataOffset - unsigned • encryptDataLength - unsigned	• -xxxxi - Invalid Init Data length • -xxxxj - Input data size exceeded the supported limits • -xxxxl - Output buffer too small • -xxxxe - Invalid key	Yes	No	Returns the data after encryption. Input data to encrypt shall not exceed 1 MB of size. Length of encrypted data is returned as 0 on any errors. For AES_CBC algorithm: • Expects IV as initData of 16 bytes in length • Null/missing IV: invalid • All-zero 16-byte IV: syntactically valid, though generally discouraged • inputData contains the buffer to be encrypted • Performs no padding. Input length must be a multiple of 16 For other algorithms, in future: Nonce can be passed as initData as well.

13	Crypto	decrypt	• algorithm - enum ◦ AES_CBC • keyId - unsigned (Reference to the loaded key) • initVectorOffset - unsigned • initVectorLength - unsigned • encryptDataOffset - unsigned • encryptDataLength - unsigned	• outputDataOffset - unsigned • outputDataLength - unsigned	• -xxxxj - Input data size exceeded the supported limits • -xxxxl - Output buffer too small • -xxxxe - Invalid key	Yes	No	Returns the decrypted clear data. Input data to decrypt shall not exceed 1.5 MB of data. length of output data is returned as 0 on any errors For AES_CBC algorithm: • Expects IV as initData of 16 bytes in length • Null/missing IV: invalid • All-zero 16-byte IV: syntactically valid, though generally discouraged • inputData contains the buffer to be decrypted.
14	Crypto	deriveSessionKeys	• algorithm - enum ◦ AUTH_DH • privateKeyId - unsigned • peerPublicKeyData - list of unsigned, ordered. • derivationKeyId - unsigned	• encryptionKeyId - unsigned • hmacKeyId - unsigned • wrappingKeyId - unsigned	None	Yes	No	Performs authenticated DH key derivation (Authenticated Diffie-Hellman)
15	Crypto	generateRandom	• Number of bytes	• randomValue - list of unsigned, ordered.	• -xxxxh - Authentication failure • -xxxxa - Operation failed	Yes	No	Returns cryptographically random data of specified length. Random value encoded is of length passed in as parameter